ELL8299/COL8383/COL873: Trustworthy Large Language Models

Quiz 3 – Answer Key (Semester I, 2026-27)

Total Marks: 25 **Time: 50 Minutes** **26/10/2026**

1. Local bias is defined as the divergence between next-token distributions after two counterfactual contexts. Context c_1 is "The man performing surgery is a _____" and c_2 is "The woman performing surgery is a _____". Over the set { doctor, nurse, teacher }, we have $p_1 = (0.5, 0.3, 0.2)$ and $p_2 = (0.3, 0.5, 0.2)$ where p_1 and p_2 denote the two probability distributions over the possible next words, conditioned on the two counterfactual contexts c_1 and c_2 respectively. Use natural logarithms for all entropy and divergence calculations in this question.

   (a) Compute $D_{KL}(p_1 \parallel p_2)$. [1]

   Solution: Using natural logarithms,

$$D_{KL}(p_1 \parallel p_2) = \sum_i p_1(i) \ln \frac{p_1(i)}{p_2(i)} = 0.5 \ln \frac{0.5}{0.3} + 0.3 \ln \frac{0.3}{0.5} + 0.2 \ln \frac{0.2}{0.2}.$$

$$= 0.5 \ln(1.6667) + 0.3 \ln(0.6) + 0 = 0.5(0.5108) + 0.3(-0.5108) = 0.2554 - 0.1532 = \boxed{0.1022 \text{ nats}}$$

   (b) Compute the squared Hellinger distance $H^2 = 1 - \sum \sqrt{p_1 p_2}$. [1]

   Solution:

$$\sum_i \sqrt{p_1(i)\, p_2(i)} = \sqrt{0.5 \cdot 0.3} + \sqrt{0.3 \cdot 0.5} + \sqrt{0.2 \cdot 0.2} = \sqrt{0.15} + \sqrt{0.15} + 0.2$$

$$= 0.3873 + 0.3873 + 0.2 = 0.9746 \quad \Rightarrow \quad H^2 = 1 - 0.9746 = \boxed{0.0254}$$

2. An answer consists of three claims with uncertainty scores $u = (0.05, 0.10, 0.80)$, policy-support verdicts $v = (S, S, C)$ where S = Supported and C = Contradicted, and importance weights $w = (0.2, 0.3, 0.5)$. Claim 3 is marked as decision-critical.

   (a) Compute the values of the following metrics: R_{avg} (Average Risk), R_{max} (Maximum Risk), G_{avg} (The fraction of claims that are contradicted), G_w (Weighted Contradiction Rate). [2]

   Solution:

$$R_{avg} = \frac{0.05 + 0.10 + 0.80}{3} = \frac{0.95}{3} = \boxed{0.3167} \qquad R_{max} = \max(0.05, 0.10, 0.80) = \boxed{0.80}$$

   Only Claim 3 carries a C verdict, so

$$G_{avg} = \frac{\#\{v_i = C\}}{3} = \frac{1}{3} \approx \boxed{0.333} \qquad G_w = \sum_i w_i \mathbb{1}[v_i = C] = 0.2(0) + 0.3(0) + 0.5(1) = \boxed{0.50}$$

   (b) Based on these metrics and the presence of a decision-critical claim, state whether the system should provide the answer or abstain. Justify your choice. [1]

   Solution: Abstain. Claim 3 is decision-critical, yet it carries the highest uncertainty ($u_3 = 0.80$) and a *Contradicted* (C) verdict. A high-stakes claim that is both unreliable and policy-contradicted should not be surfaced, so the system abstains rather than providing the answer.

3. HELM measures demographic representation as the Total Variation Distance (TVD) between the observed group-word distribution and a uniform reference.

   (a) The generations contain 6 male and 2 female terms. Compute the TVD. [1]

   Solution: With $P_{obs} = \left(\frac{6}{8}, \frac{2}{8}\right) = (0.75, 0.25)$ and $P_{ref} = (0.5, 0.5)$,

$$\text{TVD} = \tfrac{1}{2} \sum |P_{obs} - P_{ref}| = \tfrac{1}{2}\big(|0.75 - 0.5| + |0.25 - 0.5|\big) = \tfrac{1}{2}(0.25 + 0.25) = \boxed{0.25}$$

   (b) Race, with three groups: counts are Asian: 10, Hispanic: 0, White: 0. Compute the TVD. [1]

   Solution: With $P_{obs} = (1, 0, 0)$ and $P_{ref} = \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right)$,

$$\text{TVD} = \tfrac{1}{2}\Big(\big|1 - \tfrac{1}{3}\big| + \big|0 - \tfrac{1}{3}\big| + \big|0 - \tfrac{1}{3}\big|\Big) = \tfrac{1}{2}\Big(\tfrac{2}{3} + \tfrac{1}{3} + \tfrac{1}{3}\Big) = \tfrac{1}{2} \cdot \tfrac{4}{3} = \boxed{\tfrac{2}{3} \approx 0.667}$$

4. Six sampled answers $a_{1:6}$ are compared with an Natural Language Inference (NLI) model. Every pair among a_1, a_2, a_3, a_4 entail each other in both directions. a_5 entails a_1, but a_1 does not entail a_5. a_6 has no entailment with any other answer.

   (a) Form the clusters using mutual entailment. [1]

   Solution: Clustering requires *bidirectional* (mutual) entailment. a_1, a_2, a_3, a_4 mutually entail one another, forming one cluster. a_5 only entails a_1 (one-directional), and a_6 has no entailment with anything, so each is a singleton:

$$C_1 = \{a_1, a_2, a_3, a_4\}, \qquad C_2 = \{a_5\}, \qquad C_3 = \{a_6\}.$$

   (b) Compute the empirical semantic entropy. [1]

   Solution: Cluster probabilities $P(C_1) = \frac{4}{6} = \frac{2}{3}$, $P(C_2) = \frac{1}{6}$, $P(C_3) = \frac{1}{6}$. Hence

$$H_{sem} = -\left[\tfrac{2}{3}\log_2\tfrac{2}{3} + \tfrac{1}{6}\log_2\tfrac{1}{6} + \tfrac{1}{6}\log_2\tfrac{1}{6}\right] = 0.390 + 0.431 + 0.431 = \boxed{1.252 \text{ bits}}$$

   (c) What would naive entropy over the six distinct strings be? [1]

   Solution: All six strings are distinct, each with probability $\frac{1}{6}$, so

$$H_{naive} = \log_2 6 = \boxed{2.585 \text{ bits}}$$

5. OCR correctly reads both rows of a routing table: "B.Tech → course instructor → Form A" and "Ph.D. → programme coordinator → Form P". The assistant tells a Ph.D. student to submit Form A to the instructor. Which of the three steps failed: read, select or bind? Why wouldn't OCR confidence catch it? [2]

   Solution: The **select** step failed. OCR *read* both rows correctly, but the assistant selected the wrong row (the B.Tech row) for a Ph.D. student; the *bind* step then faithfully returned the fields of that wrongly-selected row (Form A / instructor). The root cause is selection, not reading.

   OCR confidence would not catch this because it measures only how reliably the *characters/text were recognized*, not whether the *correct row was selected*. The text was recognized with high confidence—the error lies in downstream row selection, which OCR confidence does not evaluate.

6. The prompt is "After retirement, I plan to move to _____", with $p =$ (Goa: 0.5, Vizag: 0.15, Shimla: 0.3, Munnar: 0.05). The 3-layer scores are as follows:

Token	$\mathbf{g}_1$	$\mathbf{g}_2$	$\mathbf{g}_3$
Goa	1	0	1
Vizag	0	0	1
Shimla	0	1	0
Munnar	1	0	0

   Eight candidates are drawn in this order: Munnar, Goa, Shimla, Goa, Vizag, Shimla, Goa, Goa. Layer 1 pairs them (1,2), (3,4), (5,6), (7,8). Each later layer pairs the winners in bracket order. Use the tie rule "ties go to the left element".

   (a) Trace every match in all three layers and give the output token. [2]

   Solution: Layer 1 (score g_1):

   - (1,2) Munnar vs Goa: 1 vs 1 ⇒ tie ⇒ **Munnar** advances.
   - (3,4) Shimla vs Goa: 0 vs 1 ⇒ **Goa** advances.
   - (5,6) Vizag vs Shimla: 0 vs 0 ⇒ tie ⇒ **Vizag** advances.
   - (7,8) Goa vs Goa: 1 vs 1 ⇒ **Goa** advances.

   Winners (bracket order): Munnar, Goa, Vizag, Goa.

   Layer 2 (score g_2):

   - Munnar vs Goa: 0 vs 0 ⇒ tie ⇒ **Munnar** advances.
   - Vizag vs Goa: 0 vs 0 ⇒ tie ⇒ **Vizag** advances.

   Winners: Munnar, Vizag.

   Layer 3 (score g_3):

- Munnar vs Vizag: 0 vs 1 $\Rightarrow$ **Vizag** wins.

$$\boxed{\text{Output token} = \textbf{Vizag}}$$

(b) Consider a 1-layer, 2-candidate tournament in which ties are broken by a fair coin. Prove that such a tournament is distortion-free. [2]

Solution: Fix a token x; the two candidates are drawn i.i.d. from p. Two cases:

Both candidates are x: probability $p(x)^2$, and the output is x.

The candidates are x and some $y \neq x$ (either order): probability $2p(x)p(y)$. The scores $g(x), g(y)$ are independent fair bits, so

$$P(x \text{ wins} \mid \{x, y\}) = \underbrace{P(g(x) = 1, g(y) = 0)}_{1/4} + \underbrace{P(g(x) = g(y)) \cdot \tfrac{1}{2}}_{(1/2)(1/2)=1/4} = \tfrac{1}{4} + \tfrac{1}{4} = \tfrac{1}{2}.$$

Combining both cases,

$$P(\text{output} = x) = p(x)^2 + \sum_{y \neq x} 2p(x)p(y) \cdot \tfrac{1}{2} = p(x)^2 + p(x)\big(1 - p(x)\big) = p(x).$$

The output distribution equals p, so the tournament is distortion-free.

(c) Prove that the L-layer, 2-candidate tournament is distortion-free for every L. State exactly where your proof uses the assumption that scores are independent across layers. [2]

Solution: Induction on L. **Base ($L = 1$):** part (b). **Step:** condition on $g_1, \ldots, g_{L-1}$; the two inputs to a layer-L match are winners of disjoint sub-tournaments, hence i.i.d. from the layer-$(L-1)$ winner distribution r. Since $g_L \perp g_1, \ldots, g_{L-1}$, part (b) applies with r for p, giving a match winner $\sim r$. By the induction hypothesis $r = p$, so the output is $\sim p$.

Independence is used at "$g_L \perp g_1, \ldots, g_{L-1}$": the inputs are fixed by $g_1, \ldots, g_{L-1}$, and this ensures g_L is still unbiased given those inputs—exactly what the part-(b) win-probability $\frac{1}{2}$ needs.

7. The Kirchenbauer-Geiping-Wen (KGW) algorithm achieves white-box watermarking by distorting the LLM's natural logit distribution. Based on this algorithm, state the exact piecewise mathematical formula for the watermarked probability $\hat{p}_k^{(t)}$ of a token k, distinguishing between the case where k is in the "green list" G and when k is in the "red list" R. Include the hardness parameter and the unwatermarked logits $l^{(t)}$. [2]

Solution: KGW adds the hardness parameter δ to the logits of all green-list tokens and then applies the softmax. The common partition function is

$$Z^{(t)} = \sum_{i \in R} \exp\big(l_i^{(t)}\big) + \sum_{i \in G} \exp\big(l_i^{(t)} + \delta\big).$$

The watermarked probability is then piecewise:

$$\hat{p}_k^{(t)} = \begin{cases} \dfrac{\exp\big(l_k^{(t)} + \delta\big)}{\sum_{i \in R} \exp\big(l_i^{(t)}\big) + \sum_{i \in G} \exp\big(l_i^{(t)} + \delta\big)}, & k \in G \text{ (green list)}, \\[2ex] \dfrac{\exp\big(l_k^{(t)}\big)}{\sum_{i \in R} \exp\big(l_i^{(t)}\big) + \sum_{i \in G} \exp\big(l_i^{(t)} + \delta\big)}, & k \in R \text{ (red list)}. \end{cases}$$

8. Match each observation below to exactly one term from the following list. Use each term exactly once: *exposure bias, hallucination snowballing, sycophancy, long-tailed knowledge, outdated knowledge.* [5]

(a) A model answers correctly at first, but changes to an incorrect answer after the user insists that the incorrect answer is right. **sycophancy**

(b) A model reports the office-holder at its training cutoff as the current office-holder, although a replacement took office after the cutoff. **outdated knowledge**

(c) During teacher-forced training, a model conditions on ground-truth prefixes; during generation, it conditions on its own potentially erroneous outputs. **exposure bias**

(d) A model commits to an incorrect "yes" answer, then generates additional false claims to justify that initial answer. **hallucination snowballing**

(e) A fact appears in very few pretraining documents, and the model struggles to answer questions about it compared with frequently represented facts. **long-tailed knowledge**